

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 19

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Doubling the distance from 1 m to 2 m reduces the dose rate from 400 to 100 (one quarter) • Distance $\times 10$ (1 to 10 m) reduces it by 100 times (400 to 4); dose is proportional to $1/\text{distance}^2$	2
01.2	• $400 / 20^2 = 1 \mu\text{Sv h}^{-1}$	1
01.3	• Sealed container prevents contact / contamination so exposure stops when the worker moves away; container may absorb alpha completely; remote handling • Barriers (lead, concrete, thick steel flask walls) absorb radiation before it reaches the worker; thickness depends on radiation type • Dose received is proportional to time near the source so work is organised to minimise time close to it (job rotation) • Dosimeters / film badges record exposure; air monitors detect alpha-emitting particles; contamination monitors on leaving check that no material is carried out	4
01.4	• Stable iodine saturates the thyroid • So radioactive iodine-131 released is not taken up / concentrated in the thyroid gland	2
01.5	• Exposure should be As Low As Reasonably Achievable	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 302-303; AQA spec 3.4.3.2.14.	
02.1	• C	1
02.2	• Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions	2

Question	Marking guidance	Marks
02.3	- Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify - So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
03.1	- Potable: no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints - Spray irrigation: low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food - Condenser cooling: no gross solids - pipe blockage - Boiler water: no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes	4
03.2	- Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter - Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed	2
03.3	- Clay particles carry electrostatic charges that make them repel each other so they stay suspended - Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier	2
03.4	<i>Any 2 from:</i> - Chlorine - Ozone - Ultraviolet light	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
04.1	- Molten iron from the blast furnace is converted directly to steel on the same site - So it does not cool and need re-melting; transport energy between sites is also avoided; waste heat from one process used by another (eg Kalundborg)	2
04.2	- Collecting and processing small dispersed quantities can use more energy in transport than is saved - Mass production and large-scale mining have economies of scale; some materials (glass) take more energy to make than plastic but can be refilled many times	2
04.3	- Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared); overhead cables cheaper than underground which need cooling - ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided	4
04.4	<i>Any 2 from:</i> - Proximity to the grid / demand to reduce transmission losses and cable costs - Access to cooling water for thermal stations - Resource availability (wind, tidal range, fuel deliveries) - Land availability and planning / public acceptance; stable geology	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	
05.1	• 25 g ha ⁻¹ (accept 24-27)	1
05.2	• The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species	1
05.3	<i>Any 3 from:</i> • The MDAF (minimum dose always fatal) of the pest is about 50 g ha ⁻¹ • This is higher than the MDNF of the bee (about 20 g ha ⁻¹) • So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha ⁻¹ • A dose of 30-40 g ha ⁻¹ would kill most pests with limited bee deaths (compromise)	3
05.4	<i>Any 2 from:</i> • Spray at night / when bees are not flying • Spray when the crop is not in flower • Spray on still days to reduce spray drift • Use a systemic pesticide / seed coating rather than spraying • Use smaller droplets for more even coverage	2
05.5	<i>Any 3 from:</i> • Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected • A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) • So fewer non-target deaths / less loss of predators, pollinators • Fewer indirect (food chain) effects / fewer inter-species relationships lost • Example: pyrethroids have high insect toxicity but low mammal toxicity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	
06.1	• Effluent gases pass through a chamber containing electrically charged wires / plates • Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick • Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials)	3
06.2	• Effluent gases are forced to rotate / spin in a cylindrical chamber • The (denser) particles are thrown outwards to the chamber walls where they fall and collect; cleaned gases leave through a central pipe	2
06.3	• The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke	1
06.4	• Turbochargers deliver more air / oxygen to the combustion chamber • So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration)	2
06.5	• Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled • They penetrate deeper into the lungs / alveoli and are not cleared; may carry toxic / carcinogenic chemicals	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	

Question	Marking guidance	Marks
07.1	• $(60 \times 45) / 9$ • = 300	2
07.2	<i>Any 4 from:</i> • Marked individuals mix freely / randomly with the rest of the population • No births, deaths, immigration or emigration between samples • Marking does not affect survival or behaviour (eg make them more visible to predators) • Marks do not wear off / fall off • All individuals have the same chance of being caught (traps do not attract or repel marked animals)	4
07.3	• Fewer marked beetles would survive to be recaptured • So the proportion marked in the second sample would be too low and the population would be overestimated	2
07.4	• Enough time for marked beetles to mix with the population • Not so long that births / deaths / migration or loss of marks change the population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
08.1	• CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again • During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls • In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises • The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large	4
08.2	• Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) • So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean	2
08.3	<i>Any 2 from:</i> • Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) • Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) • Atmospheric pressure allows liquid water to exist • Distributes heat from the tropics to higher latitudes by winds • Transports water vapour to areas that would otherwise be dry	2
08.4	• Troposphere • Stratosphere	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	
09.1	• As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures)	1
09.2	• About 39 °C (accept 38-40)	1
09.3	• 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) • = 1.2 mg dm⁻³ (accept 1.1-1.3)	2

Question	Marking guidance	Marks
09.4	- Figure 1 shows the maximum / saturation values; processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line - Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster; values above the line are impossible	2
09.5	<i>Any 3 from:</i> - Increased metabolic rate so more energy used / less surplus for growth - Eggs develop faster / hatch earlier (out of step with food supply) - Non-indigenous warm-water species may thrive and out-compete indigenous species - Toxicity of other pollutants is greater as toxins are metabolised faster - Reduced resistance to disease - Proteins / enzymes denatured if temperature is very high (above about 40 °C)	3
09.6	Hot water is sprayed from a shower floor in a tall hollow tower; air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable; demand fluctuates daily, weekly, seasonally and short-term; supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants; large-scale electricity storage not yet possible directly so energy must be converted • Pumped-storage HEP: proven, rapid response, peak shaving, but limited sites • Batteries: lithium-ion for grid and vehicles; factors affecting viability; V2G • Hydrogen: electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid; high energy density enables vehicle use • Compressed air with heat recovery; thermal storage (molten salt for CSP, high-volume stores, thermal mass); flywheels; supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage; energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage; judgement	9
10.2	• Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25